

Project 2 Checkpoint

DSC 106 · World Bank Human Development Indicators, 1960–2020

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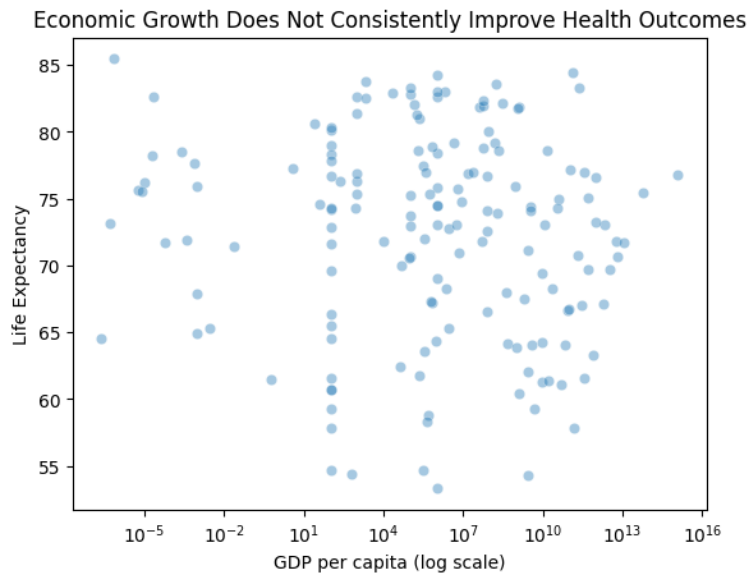
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Proposition: “Since 2000, economic growth has not consistently improved health outcomes across countries.”

Leaning: Con side (growth does improve health).

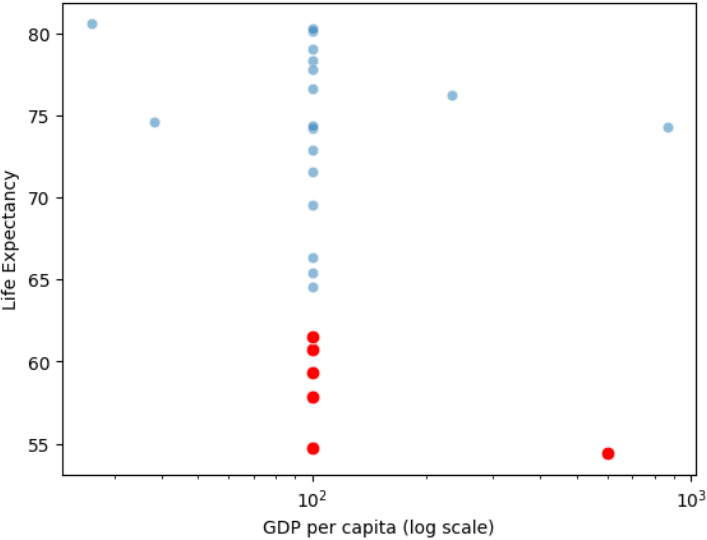
Page 1 · Pro Side (Proposition is TRUE)

Visualization A. Global scatterplot of GDP per capita (log x-axis) versus life expectancy at birth, using the most recent year available per country.



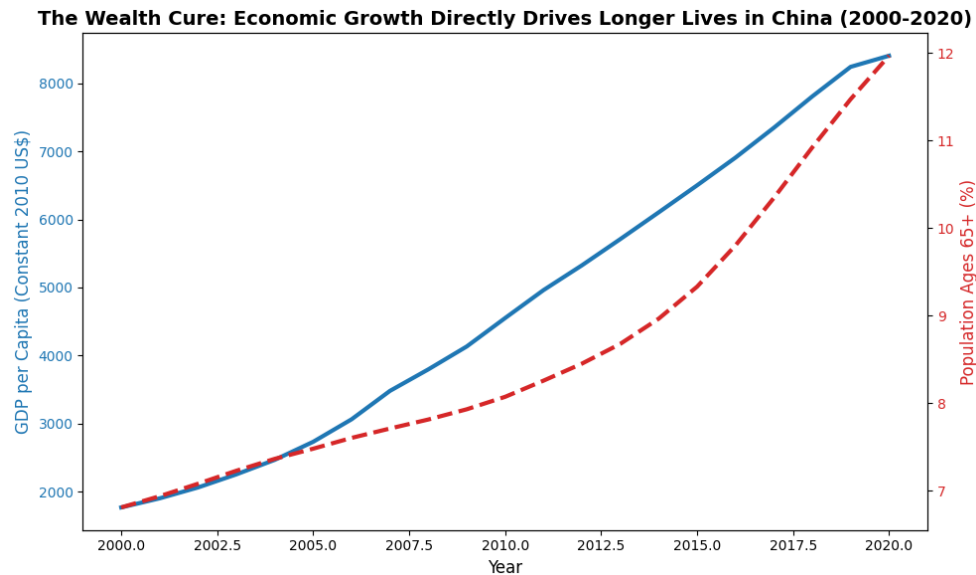
Visualization B. Same base data, filtered to a middle-income GDP band, with the seven lowest-performing countries on life expectancy highlighted in red.

Even Wealthier Countries Do Not Consistently Achieve Better Health Outcomes

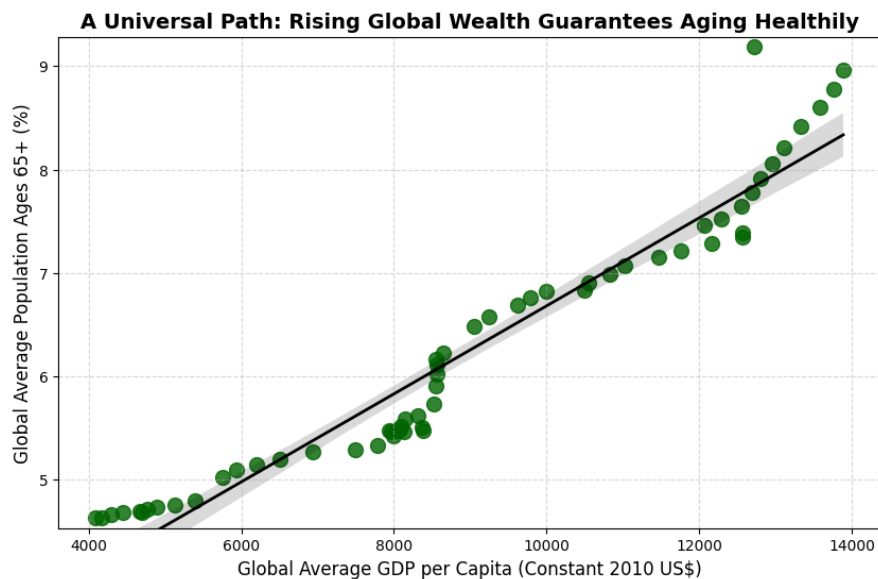


Page 2 · Con Side (Proposition is FALSE)

Visualization C. Dual-axis line chart for China since 2000. GDP per capita on the left axis (blue), share of population aged 65 and older on the right axis (red).



Visualization D. Scatterplot of global annual averages from 2000 to 2020, plotting mean GDP per capita against mean share of population aged 65 and older across all countries, with a fitted regression line.



Page 3 · Writeup

Which side we are leaning toward: We are leaning toward the **con side** (growth does improve health). The con-side visualizations read as visually tighter and harder to dismiss at a glance. They lean on widely accepted chart conventions that register as earnest even while the framing is slanted.

Pro side: proposition is TRUE

- **Slanted title.** The title frames a chart that still shows an upward trend, steering the reader to focus on spread rather than slope.
- **Log x-axis compression.** A log scale compresses the distance between rich and poor countries and makes the relationship look weaker than it is.
- **Cherry-picked highlight.** Only the seven worst-performing countries inside the filtered GDP band are marked in red.
- **Mislabeled filter.** The “wealthier countries” frame invites a stronger interpretation than the filter actually supports.
- **Uneven cross-section.** The “most recent year” is pulled independently for each country and that inconsistency is not disclosed.

Con side: proposition is FALSE

- **Dual-axis synchronization.** Two independent series are visually locked into a single story, which can make the relationship appear tighter than it is.
- **Proxy substitution.** Share of population ages 65+ is used as a health proxy even though it is also shaped by birth rates and age structure.
- **Aggregation to wash out variance.** Global averages erase country-level differences and make the trend look more universal.
- **Overstated title.** Words like “guarantees” and “directly drives” push correlation into the language of certainty.
- **Regression on aggregated means.** Fitting the line on annual global means produces a much tighter-looking relationship than country-level data.